SCOPE OF VARIABLES

When variables are created, the dict stack and the dict-tos (used to mark the top of the stack) will look at all entries from the top to the bottom to find a new variable.

I already looked at function calls and I know that when res-rip-tos is < res-rip, the program is running in '_start:' function.

I made a single reserv called last-glob-var each time we create a new variable in _start:. This points to the latest / last entry in the dict stack. Anything from this point and below is a global variable in _start:

I created another stack that stores the position of the dict where we need to scan from -> to. This is a simple way to scan only variables created in the function we are in and global variables within the function '_start:'.

```
func: function1
    1 int data-1 def-i
    100 int total def-i
    data-1 peek print-i
    total peek print-i
    glob-var peek print-i
    function2 call-func
    return
```

```
func: function2
    2 int data-1 def-i
    200 int total def-i
    data-1 peek print-i
    total peek print-i
    glob-var2 peek print-i
    function3 call-func
    return
```

```
func: function3
    3 int data-1 def-i
    300 int total def-i
    data-1 peek print-i
    total peek print-i
    glob-var peek print-i
    return
```

```
-start:
    1000 int glob-var def-i
    2000 int glob-var2 def-i
    function1 call-func
```

The diagrams on the opposite page show the state of the 'dict' stack for each function call.

All local variables in scope are coloured in green.
All local variables out of scope are colored in Red.
All Global Variables are colored in Yellow.

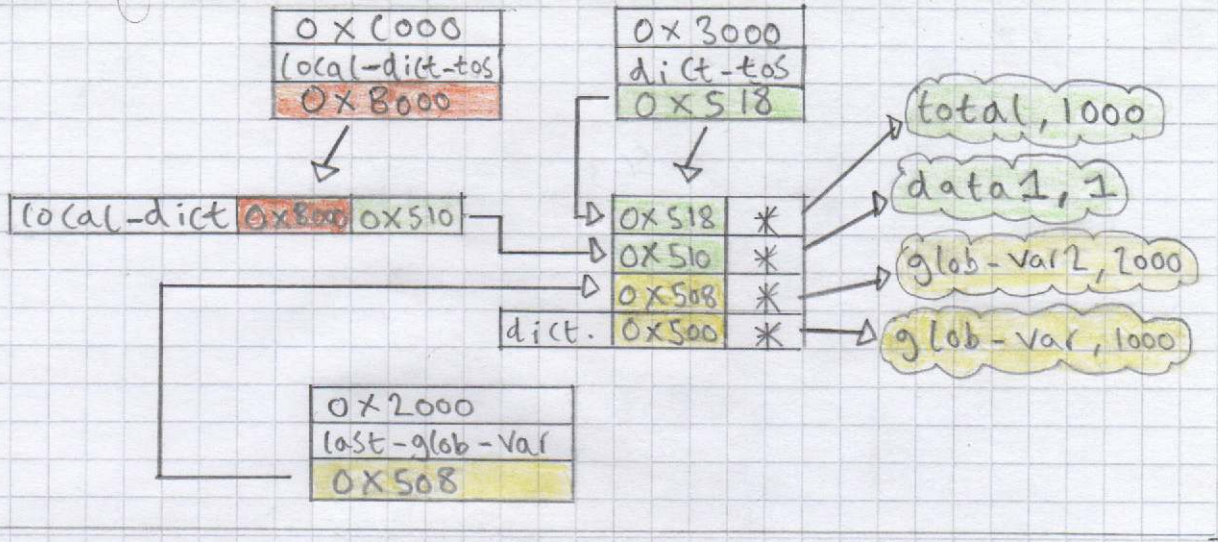
Any variables declared in -start: are global, but must not be overridden.

To see how GLOBAL variables are accessed, the variable 'last-global-var' will to point to an address up until the 'dict' stack. Any values from this 'dict' address up until the 'last-glob-var' value are GLOBAL values.

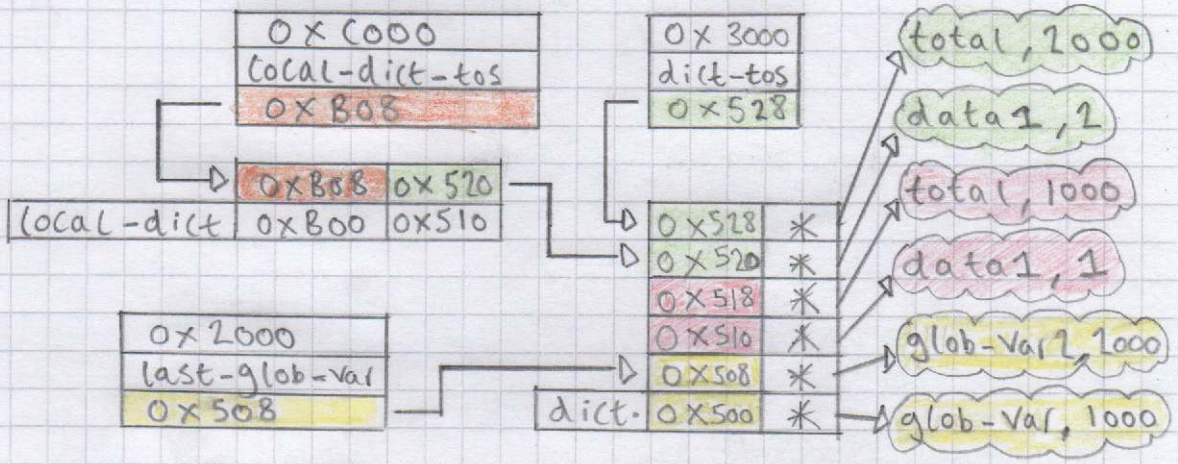
Local variables are all stored in the 'dict' stack. The 'local-dict' stack keeps track of the 'dict' stack from each function we enter. Everytime there is a function call, a new entry is added to the 'local-dict' stack. The 'local-dict-tos' points to the top of this 'local-dict' stack. When the function ends, the last entry in the 'local-dict' stack is popped off.

Look at the diagrams on the next page for each function. Local variables are easily accessed and out of scope variables are hidden. The value held by 'local-dict-tos' is the location in the 'local-dict' where it's value marks the 'dict' stack to access variables. The addresses from this address until the address held by 'dict-tos' are all in SCOPE!

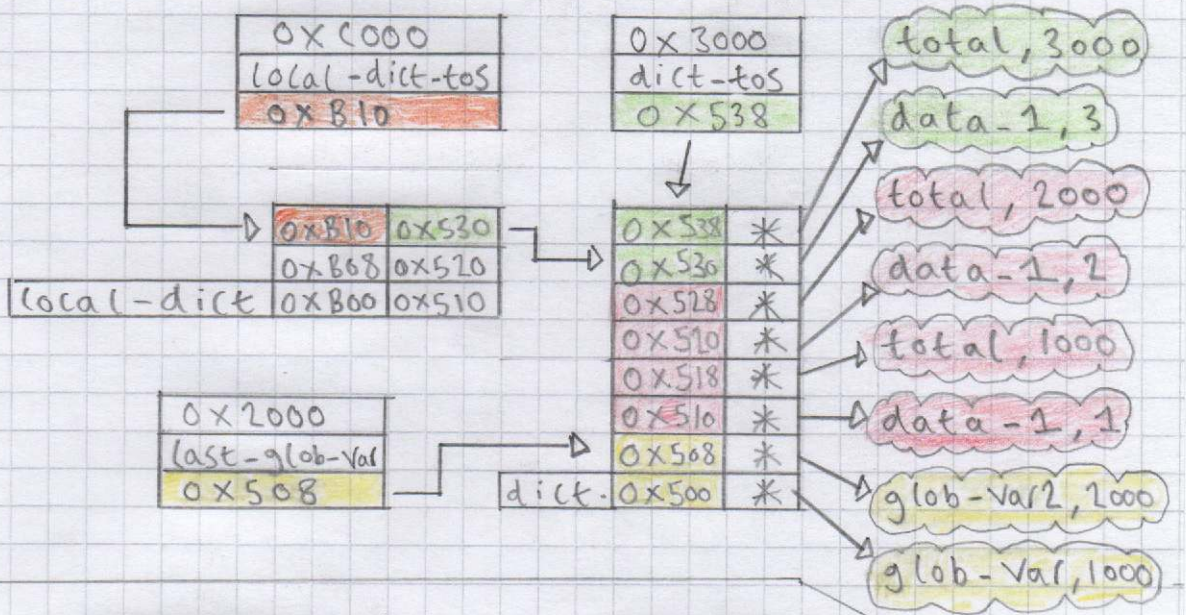
State of Variable Stacks in function 1



State of Variable Stacks in function 2



State of Variable Stacks in function 3



State of Variable Stacks in -start

